

A high-confidence causal bQTL atlas reveals the regulatory architecture of natural gene expression variation in maize

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Abstract

Binding quantitative trait loci (bQTL) identify sequence variants that alter transcription factor (TF) occupancy, but linking individual variants to causal gene expression changes remains challenging due to linkage disequilibrium and motif redundancy. Here, we apply a six-step filtering pipeline to 147,942 maize bQTL from a 25-hybrid pan-cistrome to isolate 728 high-confidence *causal* variants—positions where a single nucleotide change at a single bQTL per gene causally shifts allele-specific expression (ASE) across multiple NAM hybrids (FDR < 0.05). We demonstrate that within the actual ~160 bp MOA-seq binding peak, the disrupted TF family’s motif is the *sole copy* in 47.8% of bQTL (and 28.2% of all family-at-variant instances), with striking variation across families—BES1 (86%), Nin-like (85%), and CAMTA (70%) are almost always non-redundant, while C2H2, MYB, and ERF (~10%) maintain multiple copies even within narrow peaks. Crucially, 32.5% of variants simultaneously disrupt one TF family’s motif while creating another—with zero self-switches—revealing that causal bQTL *rewire* TF occupancy rather than simply ablate it. These 728 target chaperones (8.7%), and transcription factors (4.4%)—rather than metabolic enzymes, and are enriched for the pentose phosphate pathway (OR = 3.37), steroid biosynthesis (OR = 3.60), and ubiquitin-mediated proteolysis (OR = 2.14). Clustering by disrupted TF motif reveals that ERF and MYB each exclusively regulate 20 target genes, while all TF families converge on the same functional targets (signal transduction, protein quality control), consistent with chromatin accessibility—not TF identity—being the primary gate for functional variation. We identify 34 transcription factor genes that are themselves causal bQTL targets, creating regulatory cascades where a single upstream motif disruption can propagate to 26–96 downstream genes. This atlas provides a prioritized set of 728 single-variant regulatory switches suitable for functional validation and precision breeding.

1 Introduction

A central question in quantitative genetics is how sequence variation at cis-regulatory elements translates into phenotypic diversity. In maize, the pan-cistrome study by Engelhorn et al. [Engelhorn et al., 2025] identified 147,942 binding quantitative trait loci (bQTL)—positions where at least one of 25 F1 hybrids (B73 × diverse founders) shows allelic differences in TF binding relative to the B73 reference. These bQTL collectively explained the majority of heritable trait variation for over 70% of 143 phenotypes tested, establishing cis-regulatory variation as a dominant force in maize biology.

However, not every bQTL is equally informative. Many bQTL are in linkage disequilibrium (LD), cluster in the same promoter, or disrupt motifs that have redundant copies nearby. To

move from correlative association to mechanistic understanding, we need to identify variants where we can confidently attribute a causal chain: *this specific nucleotide* disrupts *this specific TF binding site*, which shifts expression of *this specific gene*.

Here, we describe a systematic filtering approach that distills 147,942 bQTL down to 728 high-confidence causal variants. Each passes three criteria: (1) it is the *only* significant bQTL within 2 kb of its target gene’s TSS, eliminating LD ambiguity; (2) it falls within an actual MOA-seq binding peak, confirming the variant is in actively bound chromatin; and (3) the genotype at this position causally predicts allele-specific expression (ASE) across 19 genotyped NAM hybrids (FDR < 0.05). We then characterize these 728 genes and their disrupted TF motifs to reconstruct the regulatory architecture of natural expression variation.

2 Results

2.1 A six-step pipeline identifies 728 high-confidence causal bQTL

We developed a filtering pipeline (Table 1) that progressively enriches for variants with unambiguous causal interpretation. Starting from ~237,000 MOA-seq binding peaks per hybrid, we first identified 147,942 positions with allelic binding differences (bQTL). Of these, 19,368 bQTL–gene pairs had a bQTL within 2 kb of a gene’s transcription start site (TSS) and available ASE data.

The critical step is the genotype-stratified causal test. For each bQTL–gene pair, we split the 19 genotyped NAM hybrids into two groups based on their allele at the bQTL position, then compared the magnitude of allele-specific expression (ASE) between groups using a Mann–Whitney U test. Under the null hypothesis that the bQTL has no effect on expression, the ASE distributions should be identical regardless of genotype. We observed 1,403 significant associations at FDR < 0.05—a 4.85-fold enrichment over the ~290 expected by chance ($p < 10^{-100}$, binomial test), confirming that the genotype–ASE signal is overwhelmingly real.

To avoid LD confounding, we restricted to genes with exactly one significant bQTL in their promoter (772 genes, 75% of the significant set). Finally, we required the bQTL to fall within at least one hybrid’s MOA-seq peak file, yielding the final set of 728 genes with high-confidence causal bQTL.

Table 1: **Filtering pipeline from bQTL to causal variants.**

Step	Filter	N remaining	Drop
1	All MOA binding peaks per hybrid	~237,000/hybrid	—
2	bQTL positions (allelic binding differences)	147,942	Non-variable
3	bQTL within 2 kb of gene TSS + ASE data	19,368 pairs	Distal or no ASE
4	Genotype → ASE causal test, FDR < 0.05	1,403 pairs	Non-significant
5	Single bQTL per gene (no LD ambiguity)	772 genes	257 multi-bQTL
6	bQTL within MOA-seq peak	728 genes	44 not in peak

2.2 Linkage disequilibrium and the single-bQTL design

Among the 1,403 significant bQTL–gene pairs, 75% of genes have exactly one significant bQTL, presenting no linkage concern. For the remaining 25% with multiple bQTL per gene, 62% of within-gene variant pairs have identical genotype vectors across the 19 founders (complete LD), making it impossible to distinguish which variant is causal. Our single-bQTL filter removes this ambiguity at the cost of excluding some real signals. This conservative design ensures that every gene in the 728 set has an unambiguous causal variant.

2.3 Within binding peaks, disrupted motifs are non-redundant

A key concern with single-nucleotide regulatory variants is apparent redundancy: if a promoter contains multiple copies of a TF motif, disrupting one should have little effect. We addressed this by scanning the actual MOA-seq binding peak sequence (extracted from the B73 reference genome) for each of the 728 bQTL against all 259 PWMs from PlantTFDB 5.0 (36 TF families, log-odds threshold = 6.0). For each bQTL, we identified the tightest peak containing the variant across all 25 hybrids, then counted how many copies of each disrupted TF family’s motif exist elsewhere within that peak (Table 2).

Of 728 bQTL, 600 (82.4%) have at least one identifiable TF family motif at the variant position. Among these, **47.8%** (287/600) have at least one disrupted family where the variant-overlapping motif is the *sole copy* of that family within the peak, and 13.0% (78/600) have *all* disrupted families as sole copy. At the per-family-instance level, 28.2% (459/1,627) of all family-at-variant instances are sole copies.

Critically, sole-copy rates vary dramatically across TF families. Families with longer, more information-rich motifs—AP2 (100%), RAV (100%), BES1 (86%), Nin-like (85%), CAMTA (70%)—are almost always non-redundant within the ~160 bp peak. In contrast, families with short, degenerate motifs—C2H2 (10%), ERF (10%), MYB (9%), Dof (11%)—maintain multiple copies even within narrow peaks. This family-level variation suggests two distinct mechanisms: for high-specificity families, the bQTL eliminates the only binding site; for low-specificity families, the bQTL may reduce total binding affinity or shift the precise footprint position without fully abolishing binding.

Importantly, sole-copy and redundant bQTL show comparable effect sizes (median $|d| = 2.31$ vs 2.44, Mann–Whitney $p = 0.15$), indicating that even redundant motif disruptions produce large ASE effects—consistent with quantitative modulation of binding rather than binary on/off switching.

Table 2: **Motif redundancy within actual MOA-seq binding peaks.** Sole-copy rates vary by TF family, with high-specificity families (BES1, Nin-like) almost always non-redundant and low-specificity families (C2H2, MYB) maintaining multiple copies even within ~160 bp peaks.

TF Family	At variant	Sole copy	%
AP2	8	8	100
RAV	6	6	100
BES1	7	6	86
Nin-like	27	23	85
CAMTA	27	19	70
HSF	9	6	67
ARR-B	18	11	61
bHLH	44	19	43
LBD	51	21	41
NAC	94	32	34
G2-like	65	20	31
MIKC-MADS	92	24	26
TALE	92	16	17
ERF	123	12	10
C2H2	136	14	10
MYB	125	11	9
All families	1,627	459	28.2

2.4 Variants do not just disrupt motifs—they also create new ones

A common assumption is that regulatory variants primarily *disrupt* existing TF binding sites. To test this, we performed a bidirectional motif analysis: for each bQTL, we scanned both the B73 reference sequence and the NAM variant sequence (± 28 bp around the variant) against all 259 PWMs and classified each motif as disrupted (present in ref only), created (present in alt only), or unchanged (Table 3, Figure 1).

Of 728 bQTL, 618 (84.9%) alter at least one TF motif. Strikingly, only 24.2% exclusively disrupt motifs; **21.2% exclusively create new motifs**, and **32.5% simultaneously disrupt one family’s motif while creating another**. Effect sizes are indistinguishable across all three categories (median $|d| = 2.38$ – 2.44 , Figure 2A), confirming that motif creation produces expression changes of equal magnitude to disruption.

At the family level, the net direction varies: NAC (−27), ERF (−20), and TALE (−19) are strongly disruption-biased, while EIL (+11), WRKY (+6), and YABBY (+4) show net creation. MYB is perfectly balanced (53 disrupted, 53 created). Zero self-switches were observed—when a variant disrupts a family’s motif, it never recreates a motif for the same family, confirming that rewiring always involves a *different* TF (Figure 1A). The most frequent transitions include C2H2 $\rightarrow$ TALE (6 bQTL), SBP $\rightarrow$ C2H2 (5), MYB-related $\rightarrow$ MYB (5), and NAC $\rightarrow$ ERF (4).

Are rewired motifs sole copies within the peak? To determine whether rewiring represents a true regulatory switch rather than a change among redundant copies, we repeated the analysis within actual MOA-seq binding peaks using both the B73 and NAM allele sequences. Among 233 rewired bQTL, 22 (9.4%) are *clean switches*: all disrupted families are sole copy in the reference peak, and all created families are sole copy in the variant peak (Figure 2B). A further 30 (12.9%) have at least one sole-copy family on each side. Clean switches and partial switches show comparable effect sizes (median $|d| = 2.33$ vs 2.42 , $p = 0.57$), suggesting that even partial rewiring is functionally impactful.

Examples of clean sole-copy switches. Among the 22 clean switches, several involve genes at regulatory nodes:

- **CAMTA $\rightarrow$ ERF** at phosphofructokinase (Zm00001eb048390, $|d| = 5.29$): the sole CAMTA-family motif in the B73 peak is replaced by the sole ERF-family motif in the NAM peak, with the largest effect size among clean switches.
- **NAC $\rightarrow$ TALE/WRKY** at a proline-rich receptor kinase (Zm00001eb202030, $|d| = 2.62$, drought DOWN): a sole NAC motif replaced by sole TALE and WRKY motifs at a drought-repressed signaling receptor.
- **HSF $\rightarrow$ HD-ZIP/WOX/YABBY** (Zm00001eb371180, $|d| = 2.42$): the sole HSF motif replaced by three developmental TF family motifs in the variant peak.

Whether these switches redirect regulation between specific transcriptional programs (e.g., stress versus development) or primarily alter local binding affinity cannot be resolved from bulk tissue data (see Discussion). A threshold sensitivity analysis (Methods) confirms that both the zero self-switch finding and the equal effect sizes across categories are robust across log-odds thresholds from 4.0 to 8.0.

2.5 Variant distribution across NAM hybrids

The 728 causal variants are distributed across 19 genotyped NAM hybrids with striking uniformity (Figure 3). Each variant is carried by a median of 7 hybrids (range: 3–16), with the distribution peaking at intermediate frequencies (5–6 hybrids). No variant is fixed in all 19 hybrids, indicating

Table 3: **Bidirectional motif changes at 728 causal bQTL**. 32.5% of variants simultaneously disrupt one TF family’s motif and create another, rewiring rather than simply ablating TF occupancy.

Category	N	%	Median $ d $
Only disrupted	173	24.2	2.44
Only created	152	21.2	2.42
Both (rewired)	233	32.5	2.38
Neither	158	22.1	2.36
Total	716	100	2.40

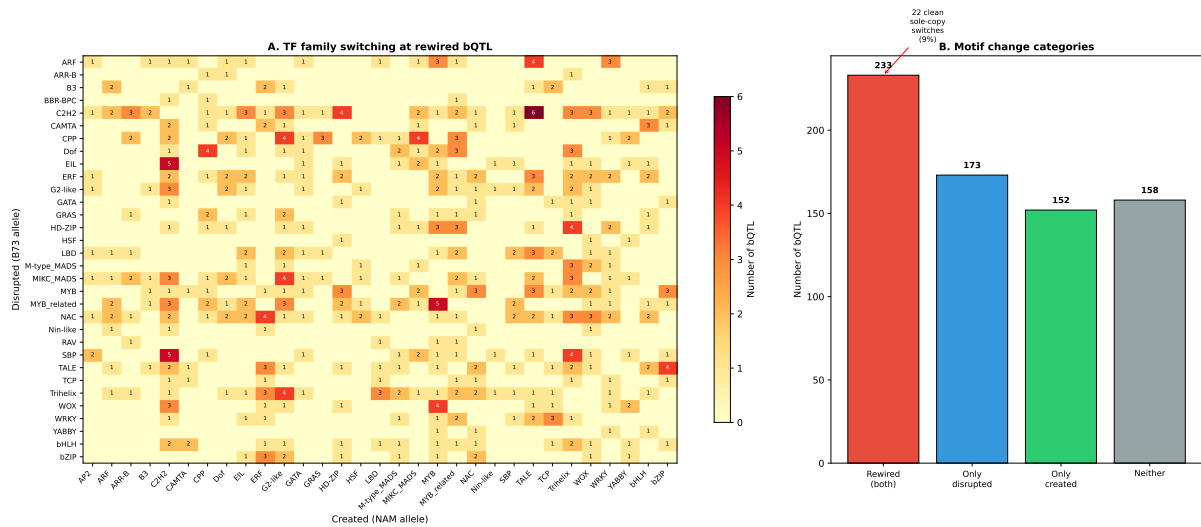


Figure 1: **TF family rewiring at causal bQTL**. (A) Transition heatmap: rows show disrupted TF families (B73 allele), columns show created families (NAM allele). No self-switches occur. C2H2 → TALE and SBP → C2H2 are the most frequent transitions. (B) Category breakdown: rewired bQTL (32.5%) are the largest category, with 22 clean sole-copy switches annotated.

that all causal variants are polymorphic in this population. Conversely, each hybrid carries 264–308 causal variants (36–42% of the 728), a remarkably narrow range suggesting that the *total regulatory variant burden* is approximately equal across diverse maize germplasm.

Rare variants (carried by ≤ 4 hybrids) show slightly larger ASE effect sizes than common variants ($\rho = -0.11$, $p = 0.003$), consistent with purifying selection acting more strongly against large-effect regulatory mutations. The effect size (Cohen’s d) for the causal test ranges from 1.0 to 14.4 (median 2.4), reflecting substantial allelic expression differences.

2.6 Target genes encode the regulatory layer, not metabolic enzymes

Functional characterization of the 728 target genes reveals a clear bias toward regulatory and signaling functions (Figure 4A): kinases/phosphorylation (76 genes, 10.4%), chaperones/HSPs (63, 8.7%), biosynthetic enzymes (39, 5.4%), transcription factors (32, 4.4%), translation machinery (29, 4.0%), signal transduction (26, 3.6%), and transporters (17, 2.3%).

Pathway enrichment analysis (Fisher’s exact test against genome background) identified five significantly enriched KEGG pathways: pentose phosphate (OR = 3.37, $p = 0.012$), steroid biosynthesis (OR = 3.60, $p = 0.016$), amino sugar/nucleotide sugar metabolism (OR = 2.11, $p = 0.021$), ubiquitin-mediated proteolysis (OR = 2.14, $p = 0.015$), and pantothenate/CoA biosynthesis (OR = 4.32, $p = 0.018$). Enriched GO terms include pentose metabolic process (OR = 19.2,

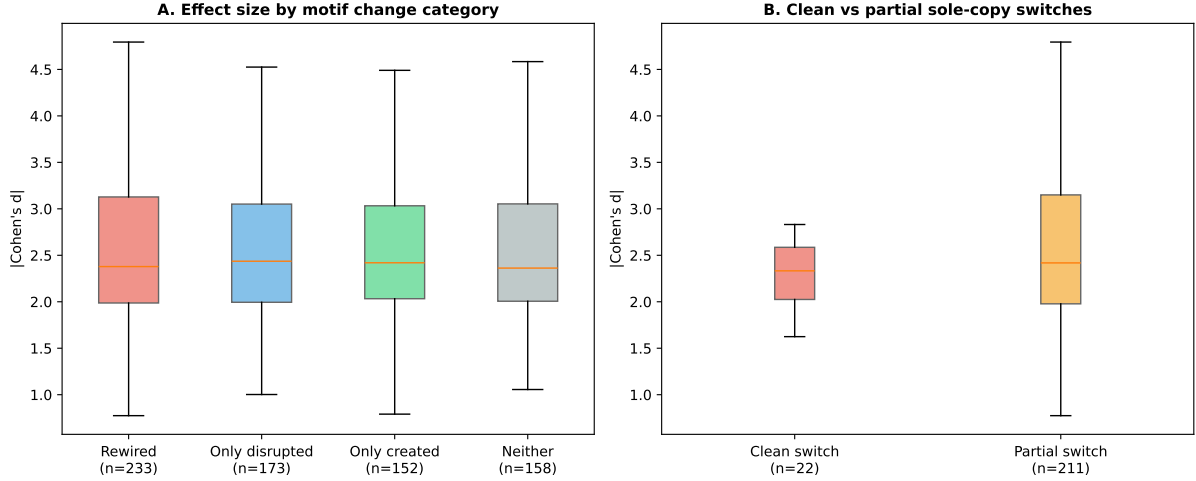


Figure 2: **Effect sizes are independent of motif change category.** (A) All four categories show comparable effect sizes (median $|d| \approx 2.4$). (B) Among rewired bQTL, clean sole-copy switches ($n = 22$) and partial switches ($n = 211$) produce indistinguishable expression effects.

$p = 1.6 \times 10^{-4}$) and regulation of hormone biosynthesis ($OR = 5.1$, $p = 0.011$).

Notably, large metabolic gene clusters (benzoxazinoid, terpene synthase) are largely absent from the 728 set—only Bx9 (UDP-glucosyltransferase) and a single flavonoid 3'-monooxygenase are present. This confirms that natural cis-regulatory variation predominantly targets the *regulatory layer*: the variant does not change an enzyme directly, but modulates how much a regulatory gene is expressed from one allele, which cascades to downstream effects.

We classified all 728 proteins into cellular functional categories (Figure 4): signal transduction ($n = 129$) is the largest, followed by protein quality control ($n = 64$), metabolism ($n = 50$), translation ($n = 31$), and transcription factors ($n = 28$). Receptors/sensors ($n = 21$), membrane/transport ($n = 24$), and hormone signaling ($n = 7$) complete the picture. Effect sizes are comparable across all layers (median $|d| = 2.4$ – 2.9), indicating no preferential impact on any particular level of the signaling hierarchy.

2.7 Gene $\times$ TF family heatmap reveals co-regulation structure

To map which TF family motifs are disrupted at each gene's bQTL, we scanned all 728 variant positions against 259 PWMs. Of 728 genes, 662 (90.9%) have at least one identifiable TF family motif at the variant position, covering 35 families. The most frequently disrupted families are C2H2 and MYB (each 160 genes), ERF (151), NAC (118), TALE (115), and MIKC-MADS (107).

Hierarchical clustering of the gene $\times$ TF family binary matrix (Figure 5) reveals that most genes have 2–5 TF families disrupted at their bQTL position, reflecting the overlapping nature of TF binding motifs. Only 134 genes (18.4%) have a single exclusive TF family disrupted. Among these, ERF and MYB each exclusively control 20 target genes, followed by NAC (9), Dof (8), and SBP (7).

The Jaccard overlap between TF family target sets (Figure 6B) shows moderate overlap ($J = 0.2$ – 0.4), confirming that the same genes often have motifs for multiple families disrupted at a single variant position. This is consistent with the dense packing of TF motifs in regulatory regions and the degenerate nature of PWM matching.

2.8 Co-regulation patterns and TF family convergence

Despite the diversity of disrupted TF families, all families regulate strikingly similar target gene profiles (Figure 4B). The top target functions for every major TF family are dominated by signal

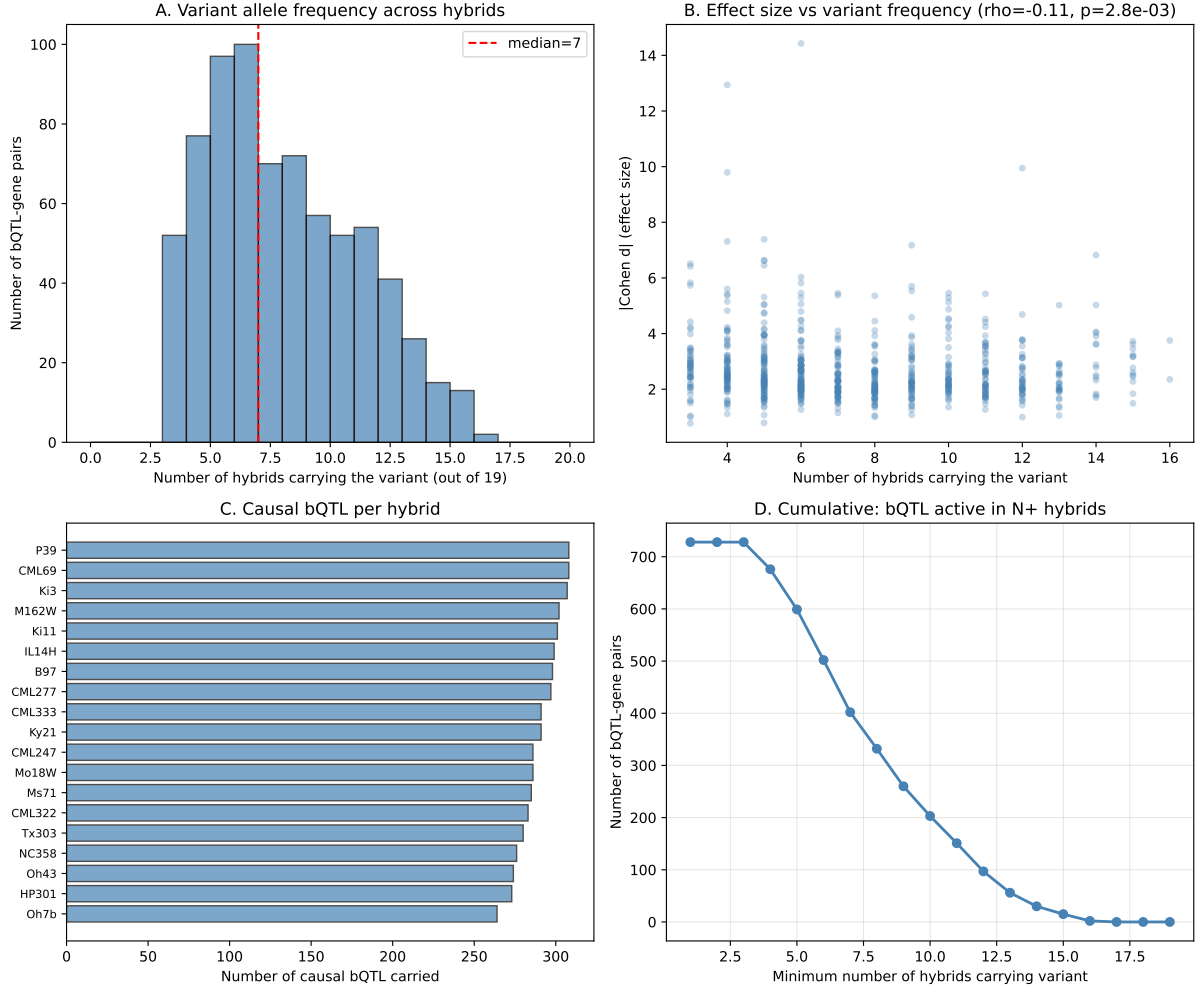


Figure 3: Distribution of 728 causal bQTL across NAM hybrids. (A) Histogram of the number of hybrids carrying each variant. (B) ASE effect size ($|Cohen's d|$) versus variant frequency, with weak negative correlation (Spearman $\rho = -0.11$, $p = 0.003$). (C) Number of causal variants per hybrid (264–308 per hybrid). (D) Cumulative distribution of variants by number of carriers.

transduction, protein quality control, and transcription factors. No family is enriched for a specific functional category (Figure 4B).

The coefficient of variation (CV) in mean effect size across TF families is only 5.5%, confirming that TF identity has minimal influence on the magnitude of allelic expression differences. This convergence supports the model that *chromatin accessibility*—whether the variant is in open chromatin and physically accessible to any TF—is the primary determinant of functional impact, not which specific TF family’s motif is disrupted [Engelhorn et al., 2025].

2.9 TF cascade network: 34 transcription factor targets amplify regulatory variation

Among the 728 target genes, 34 are themselves transcription factors, including MYB30, MYB44, WRKY50, WRKY51, NAC67, NAC82, bHLH68, bHLH112, and bHLH128. These create regulatory cascades where a single upstream motif disruption can propagate to many downstream genes (Figure 7).

The most powerful cascade nodes are:

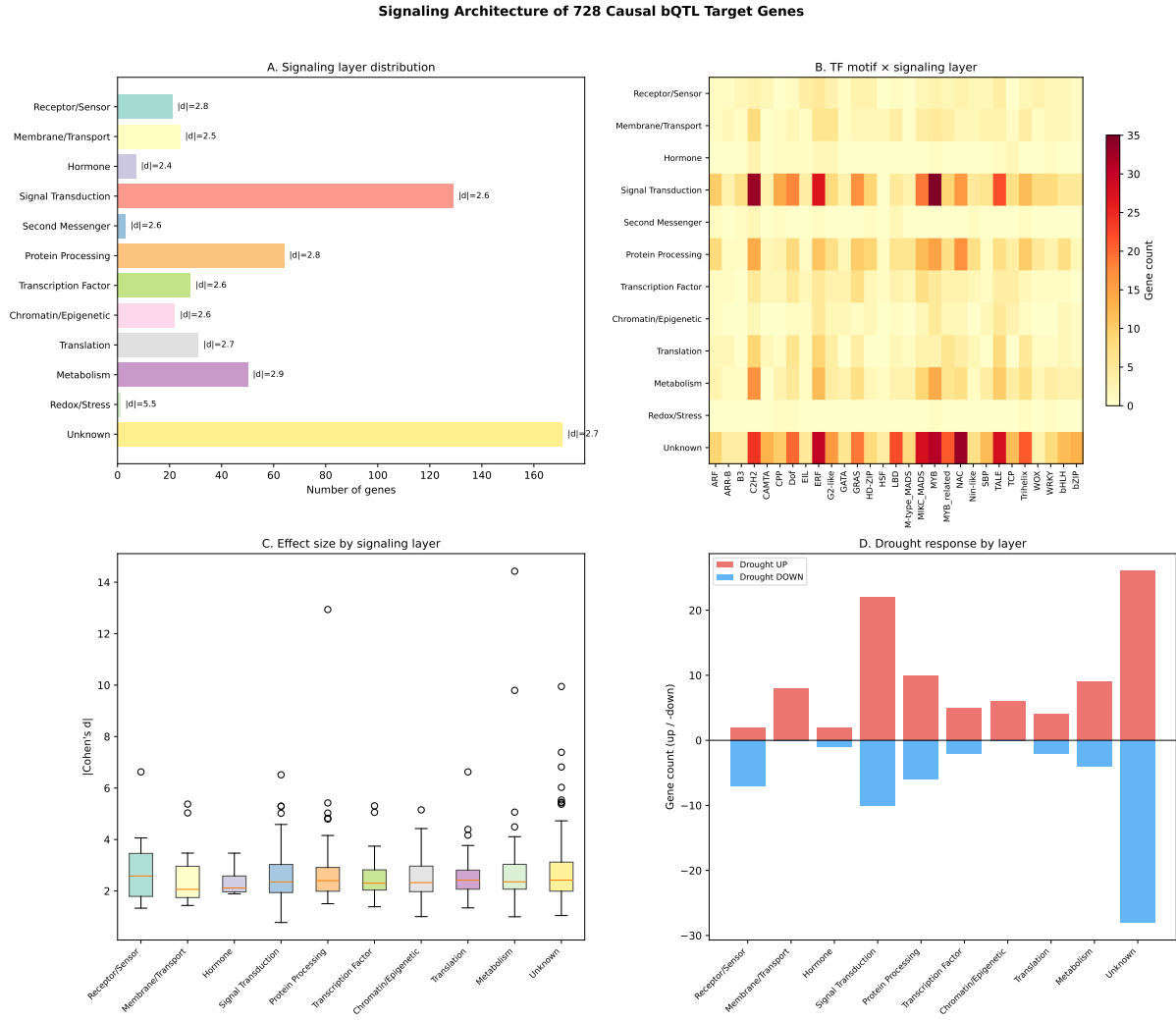


Figure 4: Functional architecture of 728 causal bQTL target genes. (A) Distribution across functional categories, with mean effect size annotated. (B) TF motif $\times$ functional category heatmap showing that all TF families regulate targets across all layers. (C) Effect size distributions are comparable across functional categories. (D) Drought response ($|\log_2FC| > 1$) by functional category; signal transduction and protein quality control show the strongest response.

- **MYB30** ($|d| = 5.31$, drought DOWN): disrupted by a MYB motif variant, MYB30 regulates 96 downstream genes in the 728 set. Its strong drought repression suggests it normally promotes growth-related targets that are silenced under stress.
- **MYB44** ($|d| = 2.28$, drought UP): disrupted by a GATA motif variant, regulates 96 downstream genes. Its drought induction and known role in ABA signaling make it a key stress-responsive cascade node.
- **WRKY51** ($|d| = 2.03$, drought UP): disrupted by GATA/HD-ZIP/WOX/YABBY motifs, a known stress-responsive TF.
- **bHLH112** ($|d| = 2.51$, drought UP): disrupted by a TALE motif, involved in drought and salt tolerance.

These cascade TFs have opposing drought responses (MYB30 DOWN vs MYB44 UP), suggesting that natural variants in their promoters can rewire the drought response by simultaneously reducing one branch of the regulatory network while activating another.

2.10 Drought integration: 27% of target genes are drought-responsive

Cross-referencing with drought expression data from [Engelhorn et al. \[2025\]](#) reveals that 16.5% of target genes are drought-upregulated ($\log_2\text{FC} > 1$) and 10.7% are drought-downregulated ($\log_2\text{FC} < -1$), with a slight overall induction bias (mean $\log_2\text{FC} = +0.206$). Among functional categories, signal transduction and protein quality control show the strongest drought response asymmetry (Figure 4D).

Notable drought-responsive targets include:

- **CIPK31** (CBL-interacting protein kinase): $\log_2\text{FC} = +1.38$, $|d| = 4.11$ —a calcium-signaling kinase with one of the largest effect sizes among drought-responsive genes.
- **PYR1** (ABA receptor): $|d| = 2.44$ —the bQTL variant affects expression of the core ABA perception machinery.
- **ARF7** (auxin response factor): $|d| = 3.47$ —linking cis-regulatory variation to auxin-mediated growth responses.
- **Phosphoenolpyruvate carboxylase 1**: $\log_2\text{FC} = +3.58$, $|d| = 2.48$ —the most drought-induced metabolic gene, connecting carbon metabolism to stress adaptation.

These examples illustrate how causal bQTL can be traced from a single nucleotide change through TF binding disruption to altered expression of defined stress signaling components.

2.11 Known signaling modules within the 728 set

Several characterized plant signaling pathways are represented by multiple members in the 728 set:

Auxin signaling. Five genes including IAA5 (auxin-responsive protein), ARF7 (auxin response factor), and auxin-induced protein 15A are regulated by variants disrupting C2H2, TALE, Dof, and TCP motifs.

ABA signaling. Fourteen genes including the PYR1 ABA receptor, regulated by variants in ERF, TCP, and CPP motifs. This module connects genotype to the core drought perception pathway.

HSP/chaperone network. Forty-eight genes form the largest coherent module, with disrupted motifs spanning many TF families. This convergence suggests that chaperone expression is broadly sensitive to regulatory variation, consistent with chaperones acting as phenotypic buffers [\[Rutherford and Lindquist, 1998\]](#).

Ubiquitin–proteasome system. Thirty-two genes involved in protein turnover, enriched in the 728 set ($\text{OR} = 2.14$, $p = 0.015$). Targeted protein degradation is a central mechanism for rapid signal transduction in plants.

3 Discussion

Our analysis of 728 high-confidence causal bQTL provides several insights into the architecture of natural gene expression variation in maize.

The regulatory layer is the primary target. The 728 causal variants overwhelmingly target signaling and regulatory proteins (kinases, TFs, chaperones, transporters) rather than metabolic enzymes or structural proteins. This is consistent with the omnigenic model [Boyle et al., 2017], where core gene expression is modulated by a large network of peripheral regulatory genes. Natural selection appears to act on the *wiring* of the regulatory network rather than on the enzymes themselves.

Chromatin accessibility gates functional variation. The finding that sole-copy rates range from 9% (MYB) to 100% (AP2) within the ~ 160 bp binding peak—yet both sole-copy and redundant bQTL produce comparable effect sizes ($|d| = 2.3$ vs 2.4)—supports a model of quantitative modulation rather than binary on/off switching. Combined with the observation that all TF families regulate similar target profiles, this suggests that the primary determinant of whether a variant “matters” is not which TF it affects, but whether the variant is in open, accessible chromatin. This echoes recent findings in human genetics where chromatin accessibility is the strongest predictor of GWAS variant functionality [Finucane et al., 2015].

Variants rewire TF occupancy, not just disrupt it. The finding that 32.5% of bQTL simultaneously disrupt one TF family’s motif and create another challenges the standard “loss-of-binding” model of regulatory variation. However, three observations suggest that the TF family *label* of the switch may be less important than the change in local binding energy: (1) effect sizes are identical across all categories (disrupted, created, rewired, neither); (2) switching to a more abundantly expressed TF family does not predict upregulation (Spearman $\rho = 0.06$, $p = 0.36$); and (3) growth/developmental versus stress/defense switches are symmetric (44% vs 56%, binomial $p = 0.45$). These results are consistent with a model where the variant alters the *physical properties* of the binding site (DNA shape, flexibility, binding free energy) in a way that affects occupancy by whichever TF happens to be present, rather than redirecting regulation from one specific transcriptional program to another.

A critical caveat is that our data derive from bulk leaf tissue, averaging signal across mesophyll, epidermis, bundle sheath, guard cells, and vasculature. Different cell types express different TF repertoires: a CAMTA motif may be functionally decisive in guard cells (where calcium signaling dominates), while an ERF motif may matter in mesophyll (ethylene response). In bulk, each TF family is the “important” one in some subset of cells, producing the appearance that all families are equivalent. Cell-type-resolved MOA-seq or single-cell chromatin accessibility data would likely reveal cell-type-specific TF hierarchies where family identity does predict effect direction—a prediction that is now directly testable.

Regulatory cascades amplify single-variant effects. The 34 TF genes that are themselves bQTL targets create an amplification layer: a variant in MYB30’s promoter does not just change MYB30 expression, but potentially affects 96 downstream genes. Importantly, cascade TFs with opposing drought responses (MYB30 DOWN, MYB44 UP) suggest that natural variation can simultaneously activate and repress different branches of the stress response network.

Implications for precision breeding. Each of the 728 variants represents a defined molecular switch: introducing or restoring a single nucleotide quantitatively modulates TF binding within the peak, causally shifting expression with a quantified effect size (Cohen’s d). The median variant is shared by 7 hybrids, making effects testable across genetic backgrounds. Combined with the interactive gene atlas (Supplementary File S1), this provides a prioritized catalog for CRISPR-based promoter engineering, where the predicted effect direction and magnitude are known *a priori*.

Limitations. Our analysis uses 19 genotyped hybrids, limiting statistical power for rare variants. The single-bQTL design excludes 25% of significant genes with multiple bQTL, potentially missing combinatorial regulatory effects. TF family assignments depend on PWM quality, and motifs for some families may be underrepresented in the PlantTFDB 5.0 database. The phenotypic consequences of these regulatory variants cannot be resolved at the individual-gene level with $N = 19$ hybrids and will require larger populations or direct functional validation.

4 Methods

4.1 Data sources

All bQTL data derive from [Engelhorn et al. \[2025\]](#), comprising MOA-seq binding peaks and allele-specific expression measurements from 25 maize F1 hybrids (B73 $\times$ NAM founders) under well-watered conditions. Genotype data from assembled founder genomes are available for 19 of 25 hybrids (excluding A188, A619, CML103, Mo17, W22). Gene annotations use the B73 NAM5 reference genome (Zm-B73-REFERENCE-NAM-5.0), with functional annotations from EntAP and EggNOG 5.0.

4.2 Genotype-stratified causal test

For each bQTL–gene pair, we extracted the genotype (reference or variant allele) at the bQTL position across 19 hybrids. We then compared the distribution of ASE values ($\log_2(\text{B73}/\text{NAM})$ expression ratios) between the variant-carrying and reference groups using a two-sided Mann–Whitney U test. Effect sizes were quantified as Cohen’s d . P -values were corrected for multiple testing using the Benjamini–Hochberg procedure.

4.3 Motif scanning and bidirectional analysis

TF binding motifs were scanned using 259 PWMs from PlantTFDB 5.0 (36 TF families) with a log-odds threshold of 6.0. The variant-overlap constraint required motif hits to physically overlap the variant nucleotide position (± 28 bp, the maximum PWM width). Motif redundancy was assessed within actual MOA-seq binding peaks (tightest peak per bQTL across all 25 hybrids). For bidirectional analysis, both the B73 reference and NAM variant sequences were scanned; motifs present only in the reference were classified as disrupted, those present only in the variant as created.

4.4 Threshold sensitivity analysis

To assess robustness, the bidirectional motif analysis was repeated at seven log-odds thresholds (4.0, 5.0, 5.5, 6.0, 6.5, 7.0, 8.0). As expected, the rewired fraction varies with threshold (18.5%–59.9%), reflecting more permissive motif calling at lower thresholds. However, three key findings are stable across all thresholds: (1) zero self-switches at every threshold; (2) motif creation remains substantial (14.6%–21.3% of bQTL exclusively create motifs); and (3) effect sizes are indistinguishable across categories at every threshold (median $|d|$ range 2.29–2.54).

4.5 Functional category classification

All 728 genes were classified into cellular functional categories based on keyword matching against EntAP protein descriptions, EggNOG COG descriptions, UniProt GO terms, and KEGG pathway annotations. Layers include receptor/sensor, membrane/transport, signal transduction, protein quality control, transcription factor, chromatin/epigenetic, translation, metabolism, hormone signaling, and redox/stress response.

4.6 Interactive atlas

The Maize Causal bQTL Atlas (<https://mbanf.github.io/causal-bqtl-atlas/>) is an interactive HTML visualization built with D3.js v7. It includes: a force-directed layered cell diagram of all 728 genes colored by regulatory layer; rich tooltips with gene description, effect size, drought response, disrupted and created TF motifs, motif rewiring category, sole-copy status within MOA-seq peaks, and hybrid carriers; filterable by motif rewiring category (rewired, only disrupted, only created, neither), regulatory layer, disrupted and created TF family (union filter), drought category, and specific hybrid; and an optional OpenAI API integration for AI-powered biological summaries of selected gene sets.

5 Data Availability

All bQTL data are from Engelhorn et al. [2025]: SRA accession PRJNA1101486, GEO accession GSE294039. The 728-gene atlas (gene_atlas_728.csv), interactive visualization (gene_atlas_728_interactive.html) and analysis scripts are available at <https://github.com/mbanf/causal-bqtl-atlas>.

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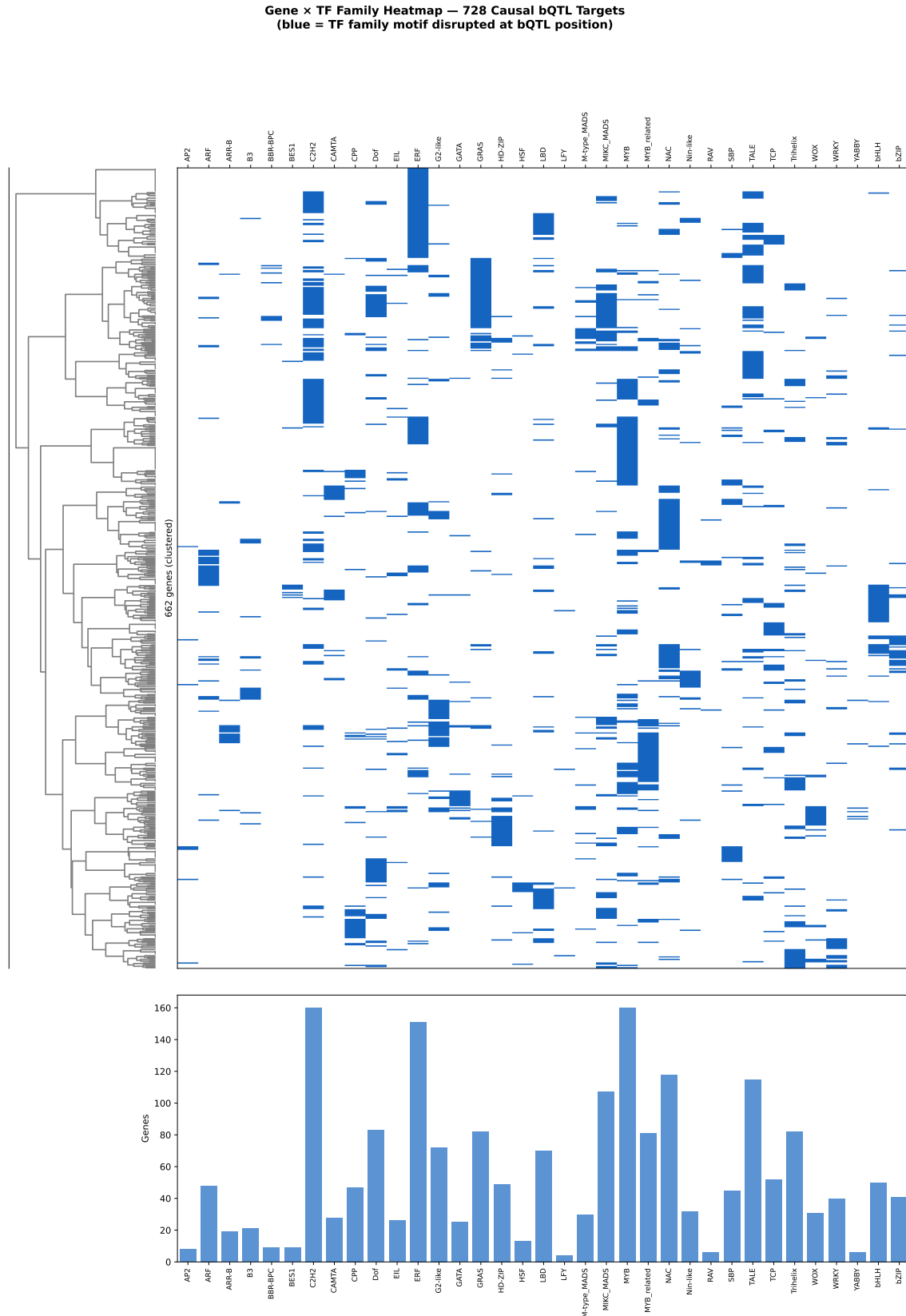


Figure 5: **Gene × TF family heatmap.** 662 genes with identifiable disrupted motifs (rows) × 35 TF families (columns). Blue indicates the TF family’s motif is disrupted at the bQTL position. Genes are hierarchically clustered by Jaccard distance; bar chart below shows total genes per family.

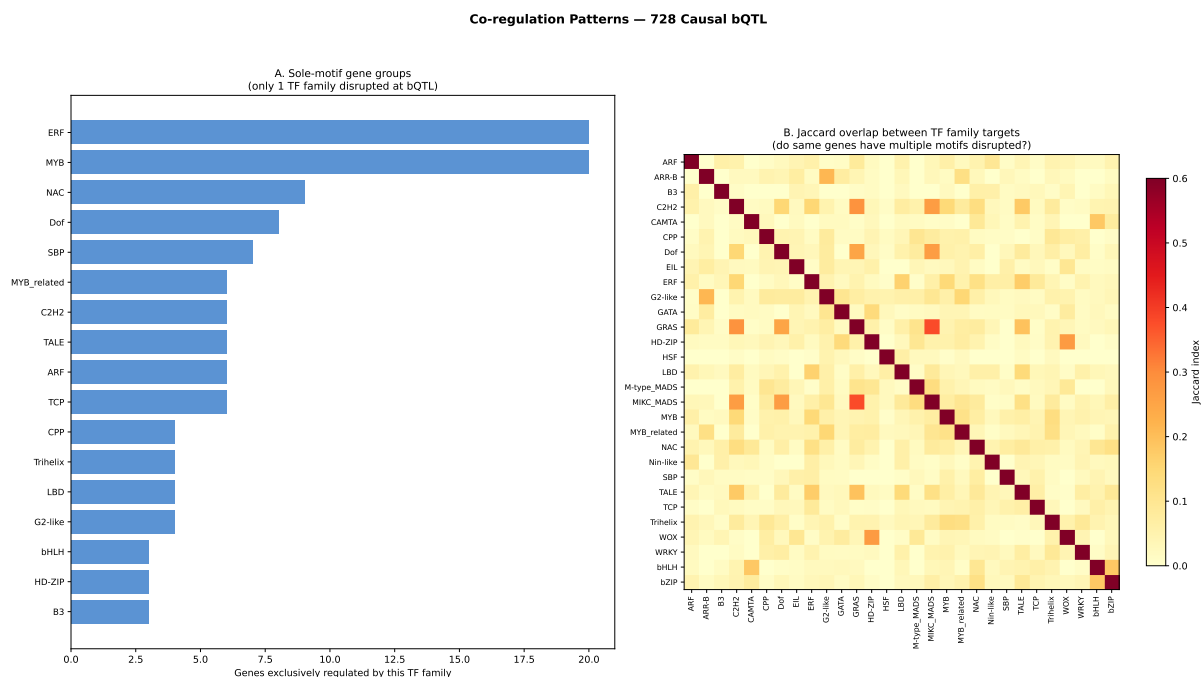


Figure 6: **Co-regulation patterns among 728 causal bQTL.** (A) Genes exclusively regulated by a single TF family (sole-motif groups). ERF and MYB each control 20 genes exclusively. (B) Jaccard similarity between TF family target gene sets. Moderate overlap ($J = 0.2\text{--}0.4$) reflects shared motif occurrence at variant sites.

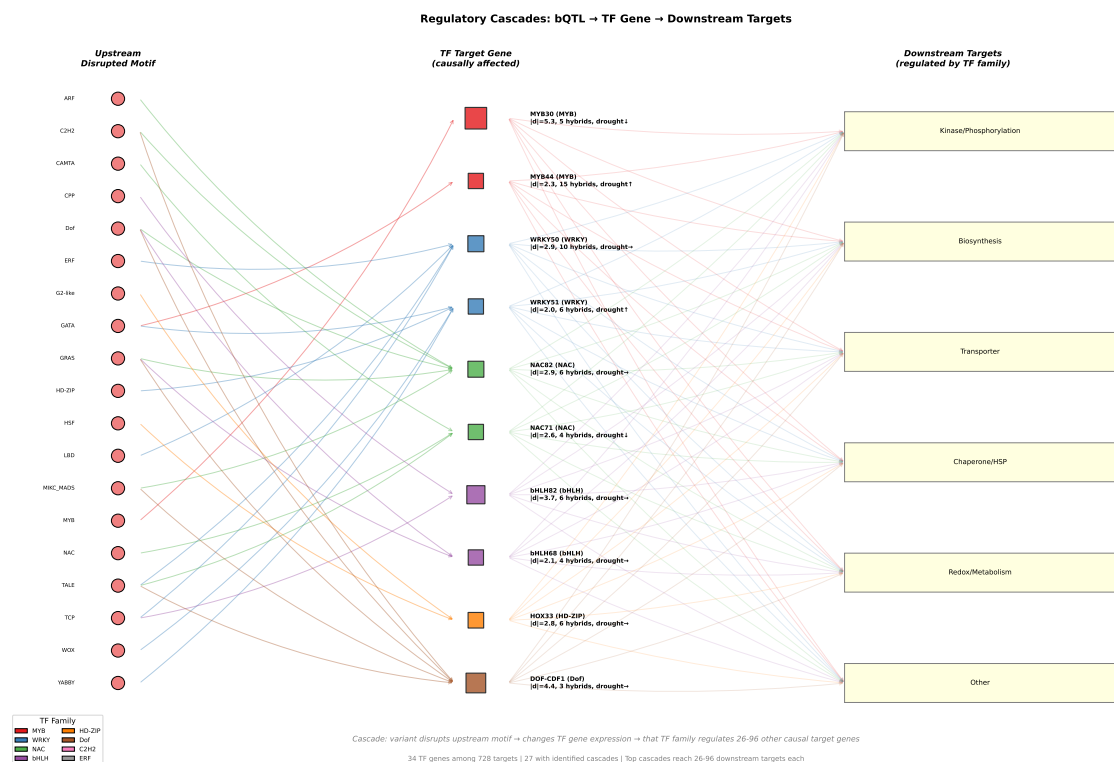


Figure 7: **TF cascade network among 728 causal bQTL targets.** 34 TF genes (middle layer) are themselves targets of bQTL that disrupt upstream motifs (top layer). Each TF in turn regulates downstream target genes (bottom layer), creating regulatory amplification cascades.